Henry Klatt

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Education

Ph.D. in Mathematics, The George Washington University 2022–present
Expected May 2027. Advisor: Valentina Harizanov.
Dissertation working title: *Effective Nonstandard Algebra*.
Specialty examination: Passed with distinction, "Forcing in Computability Theory" (Mar. 22, 2024).

M.A. in Mathematics, The George Washington University 2024

B.S. in Mathematics, Iowa State University 2022
Magna cum laude.
Undergraduate thesis: *Language, Models, Ultraproducts, and the Hyperreals* (2021).
Advisor: Tim McNicholl.

Research Appointments and Fellowships

Research Fellowship, Hausdorff Research Institute for Mathematics Fall 2025
Trimester programme: "Definability, Decidability, and Computability," Bonn, Germany.

Research Interests

Mathematical logic, primarily computability theory, with a focus on ultraproduct-like constructions in model theory and computability theory. Additional interests include knot theory, algorithmic learning theory, and number theory.

Preprints and Work in Preparation

Preprint

R. Dimitrov, V. Harizanov, **H. Klatt**, and K. Srinivasan. *On Cohesive Products of Fields* (2026).
Submitted to the *Journal of Symbolic Logic*. arXiv:2604.09965.

Work in Preparation

V. Harizanov and **H. Klatt**. *HKSS Completeness of Quandles, Racks, and Keis*.
In preparation.

V. Harizanov, M. "Turbo" Ho, **H. Klatt**, M. Kukla, and K. Srinivasan. *Cohesive Powers of Abelian Groups*.
In preparation.

V. Harizanov, **H. Klatt**, and K. Srinivasan. *Generalized Cohesive Products*.
In preparation.

W. Calvert, R. Dimitrov, V. Harizanov, **H. Klatt**, and K. Srinivasan. *Automorphisms of Effective Vector Spaces*.
In preparation.

Teaching Experience

Instructor of Record, The George Washington University Summers 2023, 2025, 2026
Math and Politics. Independently responsible for course delivery, assessment, and grading.

Graduate Teaching Assistant, The George Washington University 2022–present
Calculus I, II, and III; Math and Politics; Mathematical Ideas.

Talks

Invited Conference Talks

Title TBA (scheduled) Feb. 27–28, 2027
Southeastern Logic Symposium (SEALS), University of Florida, Gainesville, FL.

Ultraproducts of Galois Actions (scheduled) Nov. 7, 2026
AMS Fall Western Sectional Meeting, Special Session on Inverse Problems, Arizona State University, Tempe.

Cohesive Products of Galois Extensions Sept. 7–11, 2026
Computability, Complexity, and Randomness 2026, University of Leeds.

Countable Ultraproducts May 29, 2026
Logicón, Universidad Nacional Autónoma de México.

Cohesive Products of Fields Jan. 5, 2026
Joint Mathematics Meetings, ASL Special Session on Computability and Its Applications, Washington, DC.

Cohesive Powers of Number Fields Apr. 6, 2025
AMS Eastern Sectional Meeting, hosted by the University of Connecticut, Hartford.

Quandles are Complicated Aug. 1, 2024
FRG Workshop on Definability, Decidability, and Computability, Harvard University.

A maximally complicated isomorphism problem May 4–5, 2024
AMS Western Sectional Meeting, San Francisco State University.

Quandles are Complicated Dec. 10, 2023
Knots in Washington, The George Washington University.

Seminar Talks

Cohesive Products of Galois Extensions Sept. 22, 2026
Logic Seminar, Iowa State University.

Cohesive Powers of Galois Extensions Apr. 23, 2026
Online Logic Seminar.

More on cohesive products of fields, Part II Feb. 11, 2026
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

More on cohesive products of fields Feb. 4, 2026
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

Introduction to Cohesive Powers Nov. 14, 2025
Junior Seminar, Hausdorff Research Institute for Mathematics.

Types, existential closures, and cohesive powers in the study of arithmetic Sept. 2025
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

Why doesn't Chevalley's theorem solve the halting problem? Jan. 24, 2025
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

Smaller Ultraproducts, part 2 Oct. 16, 2024
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

Smaller Ultraproducts: A Survey Oct. 2, 2024
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

n-generic and more: how far can forcing take us? Mar. 27, 2024
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

The isomorphism problem for quandles is complicated Sept. 6, 2023
Logic Seminar, The George Washington University.

Infinitesimal Calculus Oct. 3, 2022
Louise Hay Logic Seminar, University of Illinois at Chicago.

Ultrapower construction of the hyperreals Sept. 15, 2022
Logic Seminar, The George Washington University.

Some fruit from set-theoretic pluralism Apr. 6, 2022
Logic Seminar, Iowa State University.

How Effective is Forcing? Nov. 30, 2021
Logic Seminar, Iowa State University.

Professional Service

Conference and Programme Organization

Computability of Algebraic Structures (scheduled) — June 20–25, 2027
Lead organizer; BIRS Workshop, Casa Matemática Oaxaca (CMO).

Special Session on Mathematical Logic — Oct. 3–4, 2026
Co-organizer; AMS Eastern Sectional Meeting, Washington, DC.

Special Session: DDC@DC — Jan. 4–7, 2026
Co-organizer; Joint Mathematics Meetings, Washington, DC.

Seminar on Computable Structure Theory — Fall 2025
Organizer; trimester programme: "Definability, Decidability, and Computability."
Hausdorff Research Institute for Mathematics.

Special Session on Computability, Complexity, and Algebraic Structure — Apr. 6–7, 2024
Co-organizer; AMS Spring Eastern Sectional Meeting, Howard University, Washington, DC.

Seminar Organization

Logic Seminar, The George Washington University — 2022–present
Co-organizer.

Reading Group Organization

Computable Structure Theory — Sept.–Dec. 2025
Co-organizer; Hausdorff Research Institute for Mathematics.

Commutative Algebra — May–July 2025
Co-organizer; The George Washington University.

Honors and Research Support

Travel Grants, Association for Symbolic Logic — 2024, 2025, 2026

Summer Research Support — 2023, 2024
Supported through advisor Valentina Harizanov's National Science Foundation grant.